

```
$ curl /users/123 | usecase | repository | db
```

Clean Architecture

// 0000000000

Controller

□□

Usecase

□□

Repository

□□

Presenter

□□□

\$ cat blueprint.md



01 /



02 /



03 /



TypeScript



04 /



DB



PART 01



□□□□□□□□



```
1 // UI 로직
2 app.get('/users', (req, res) => {
3   const users = db.query('SELECT * FROM users');
4   res.json(users.map(u => ({
5     name: u.name,
6     email: u.email,
7     createdAt: u.created_at.format('YYYY-MM-DD')
8   })));
9 });
```

INCOMING REQUESTS

GraphQL

→

→ 10

MongoDB

→ SQL

□□□□□□□□□□□□□□□□□□



□□□□□



Repository



API □□□



Presenter

□□□□□□



Controller

INVARIANT

□□□□Usecase□□□□ — □□□□□□□□□□□□□□□□

PART 02





Controller = □□□



Usecase = □□



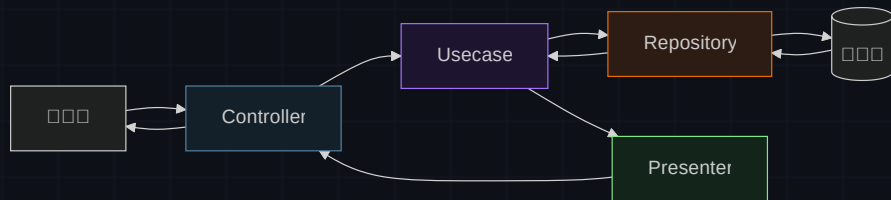
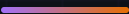
Repository = □□



Presenter = □□□



□ □ □ □ □ □ □ □ □ □



□ □

UI → Usecase → Repository → DB

□ □

DB → Repository → Usecase → Presenter → UI □ □ □ □

PART 03

TypeScript □□

□□□□□□□□□□

Repository

```
1 export interface UserRepository {
2   getUser(id: number): Promise<User | null>;
3 }
4
5 // MySQL — Usecase
6 export class MySQLUserRepository
7   implements UserRepository {
8   async getUser(id: number) {
9     const rows = await db.query(
10       'SELECT * FROM users WHERE id = ?', [id]);
11     return rows[0] ?? null;
12   }
13 }
```

Usecase

```
1 export class GetUserUseCase {
2   constructor(
3     private userRepository: UserRepository
4   ) {}
5
6   async execute(id: number) {
7     const user =
8       await this.userRepository.getUser(id);
9     if (!user) return null;
10
11     // 30 days ago
12     if (user.createdAt < thirtyDaysAgo()) {
13       return null;
14     }
15     return user;
16   }
17 }
```

Repository

Repository — MySQL MongoDB

Presenter NO GRAPH Controller

```
1 // Presenter
2 export class UserPresenter {
3   static toResponse(user: User | null) {
4     if (!user) return { error: 'User not found' };
5     return {
6       id: user.id,
7       name: user.name,
8       created_at: user.createdAt.toISOString(),
9     };
10  }
11 }
```

□□□□□□□□

```
1 // routes/user.ts — 
2 const repo = new MySQLUserRepository();
3 const usecase = new GetUserUseCase(repo);
4 const controller = new UserController(usecase);
5 
6 router.get('/users/:id', controller.getUser);
```

COMPOSITION ROOT

Controller □□□□□□□□□□ Usecase □□□□□□□□ — □□□□□□
□□□□□□

PART 04



CHECKPOINT_01

다음의 코드에서 Cache를 사용하는 예는?

A Usecase에서 Cache를 사용한다

B Controller에서 Cache를 사용한다

C Repository에서 CacheUserRepository를 사용하여 DB

FAQ

MVC

MVC Clean Architecture

10 100 1 100

2

Repository

Usecase





Layers separated.

□□□□Spring Boot × DDD□□□□□□□□□□□□

```
$ curl /users/123
```

```
controller → usecase → repository → presenter
```